

The Two Graduations

v2 · law and gauge · a spine for glide construction and cross-slit projection · every claim stated once, linked to the instrument that carries its evidence · identities re-run live at the foot of this page

0 · THE THEOREM THE TITLE NAMES

Theorem 0. A scale on a straight line cannot both mark equal steps and assign infinity a finite mark. A scale with equal steps is affine, and an affine map of the line sends ∞ to ∞ .

One line, and every depth practice must live on one side of it. The **Reach gauge** keeps equal steps and leaves infinity unmarked; the **Turn gauge** is a Möbius re-scaling that marks infinity at a finite point and pays for it with harmonic crowding. These are the two graduations. Everything below is the anatomy of that choice.

PROVED ABOVE · CROWDING AND EQUAL STEPS MEASURED AT LEDGER IDENTITIES 1-2

1 · THE SUBSTRATE

Strip both practices to what they share and one incidence skeleton remains: straight lines in the image, their concurrences, the vanishing axes, the picture plane. Call it the **armature**. Neither practice owns it. **Glide** draws it with the tinsmith's projectors; **cross-slit** draws it with light through two slits; on the drafting board they trace the same skeleton.

CHECKED AT: [SIX CORE ARMATURES](#) · [TETRA ENGINE BASE](#) · [THE MEASURING BOARD](#)

2 · LAW AND GAUGE

Two different things were hiding under one word in v1, and the theory sharpens when they are split. The **armature law** is the shape of the image machinery: each channel of the picture contracts toward its seat at its own rate – straight depth lines, a rate pair, the vanishing structure. The **depth gauge** is the scale laid along those lines – which of the two graduations, and pinned where. A **reading** is a law together with a class of gauges; a **calibration** selects the representative. So there is no single object called *the* glide: glide names the law, and the gauge remains to be chosen.

3 · THE TWO READINGS

Glide construction is the linear armature law with the gauge left free – and the practice uses that freedom. Sheet 39 runs all six core armatures on a gauge that is Möbius in depth, with affine rates laid on top; the measuring board runs the same law on an equal-step gauge and then, at the flip of a chip, on the harmonic one. Straightness survives every re-gauging because the law is affine in the gauge, whatever the gauge is in depth.

Cross-slit projection is the same skeleton with no freedom left: the optics weld law and gauge together. Rays through two slits fix both the lines' bowing and the crowding along them; no slider can repeal the bow, and no chip can re-graduate it. That is the precise content of "one is physics, the other is a tool": the tool retains a degree of freedom the optics foreclose.

CHECKED AT: SIX CORE ARMATURES (MÖBIUS GAUGE, STRAIGHT LINES – LEDGER IDENTITY 11) · A CROSS-SLIT PHOTOGRAPH · THE BOWING THEOREM · THE FUSED AND THE SPLIT

4 · THE INTERFACE

The readings meet by calibration: pin them to agree at chosen depth planes. This is not an invention of the board – it is glide practice's own **staging**, generalized. Sheet 39's gauge plane images identically under all six armatures: "the staging rectangle is the common ground every comparison stands on." The measuring board extends staging *across the two practices*: **level 0** (the front face, true size, in the picture plane) and **level 16** (the deepest constructed plane) are shared pins. Between the pins the gauges disagree in character – equal spacing against harmonic crowding. Beyond them the vanishing hierarchy opens, and it is strict: $16 < h \text{ pair} < v \text{ pair}$. Under the harmonic gauge the hierarchy sharpens into theorem: the depth plane at infinity maps to the flat segment $h_2 \cdot A \cdot h_1$ on one vanishing axis – the limit of measured depth – while the v pair lies beyond even that, unreachable at any depth, directional only.

CHECKED AT: THE MEASURING BOARD (HARMONIC CHIP) · SIX CORE ARMATURES (GAUGE PLANE) · LEDGER IDENTITIES 3-6

5 · THE TWO BALANCES

The split rectangle distinguishes two kinds of truth. Its multiplicative balance holds for

every position of the point – Turn-side truth, an invariant. Its additive balance closes *only on the diagonal* – Reach-side truth, a calibration. Both practices contain both kinds: **glide** has its ratio and parallelism invariants, **cross-slit** has its choices of screen and slits. But their characteristic truths sort by side – the cross-slit's signature laws are invariants (the cross-ratio readout that never moves, the bowing no slider can repeal), the glide's signature truths are measurements – and the *interface between readings* is purely Reach-side: shared levels are diagonal closures, Euclid I.43 moments, true exactly where the construction puts them. The cross-ratio is the balance that never moved.

CHECKED AT: TWO BALANCES IN A SPLIT RECTANGLE · LEDGER IDENTITY 10

6 · THE CLASSIFICATION

Where can the image of infinity land? The six core armatures are the answer, read off the rate pair per channel: $(1, 1)$ a point catches infinity; (λ, λ) infinity arrives early as an area; $(1-k, 1)$ the vanishing axis with its directing axis; $(0, 1)$ the herringbone – vanishing one way, parallel the other; $(0, \lambda)$ the keystone; $(0, 0)$ the box, where infinity is never marked at all. The two gut feelings sit at the corners of this table.

Reducible to a point: the $(1, 1)$ corner, reached optically by fusing the slits into a pinhole.

Broad and shallow: the $r \rightarrow 0$ corners, where recession goes parallel, depth stays a measured slab, and composition proceeds laterally by staging – one slab-like expanse of endlessly arranged armatures.

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 | Conjecture, unchecked: since the focal locus of the cross-slit congruence is a pair of lines and
 | lines are self-dual in space, the two-slit rig may sit at the self-dual center of this family, with the
 | point and slab readings as its dual ends. Flagged, not asserted.
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CHECKED AT: SIX CORE ARMATURES (THE RATE-PAIR TAXONOMY, LIVE) · THE MEASURING BOARD (PINHOLE CHIP)

7 · THE MEANS, EXACTLY

Three calibrations were tried on the way to the board, and they are tied together more tightly than a list of names suggests. Write x_g and x_{cs} for the two marks of one true measure x . The transpose calibration is the geometric mean: $x_g \cdot x_{cs} = x^2$, so inversion through x exchanges the two marks. The mirrored calibration then splits into a conjugate pair under that very inversion:

$x = AM(X_g , x^2/X_{cs})$ and $x = HM(X_{cs} , x^2/X_g)$

– the true measure is simultaneously the arithmetic mean of the glide mark with the inverted camera mark, and the harmonic mean of the camera mark with the inverted glide mark. The geometric calibration is the exchanger between the other two. The measured board completes the family: pin two levels and the gauge between them ranges over the Möbius maps, the equal-step and harmonic scales being the two distinguished members.

CHECKED AT: LEDGER IDENTITIES 7-9 · THE MEASURING BOARD

8 · VOCABULARY

TERM	LOAD IT BEARS
armature	the shared incidence skeleton – lines, concurrences, axes, picture plane
armature law	the shape of the machinery: per-channel linear contraction toward a seat at a rate; glide's identity; summarized by the rate pair
depth gauge	the scale laid along the depth lines; Reach (equal steps, ∞ unmarked) or Turn (Möbius, ∞ at a finite mark) – the two graduations of Theorem 0
reading	a law together with a class of gauges; glide and cross-slit are readings, not rival skeletons
calibration	a depth plane where readings are pinned to agree; selects the gauge's representative; glide practice's staging, generalized
invariant	what holds at every setting; Turn-side truth; the cross-ratio is the invariant every reading shares – the anharmonic interface
rate pair	the two per-channel rates; coordinates of the six-armature classification of the image of ∞
vanishing hierarchy	measured limit, then the depth plane at infinity, then the directional pair beyond reach

9 · PROVENANCE

Classical and on record: the crossed-slit image as a (1, 1) linear line congruence, with the pinhole as its point degeneration (Zomet, Feldman, Peleg & Shashua, the crossed-slits

camera, c. 2003); Klein's Erlangen program for geometry as invariants; Euclid I.43 for the additive balance; the descriptive-geometry and sheet-metal tradition for developments, transition pieces, and staging. Internal to this practice and prior to this spine: the glide system and its armatures (the author's 1984 study and the Glide_projection sheets, where the staging rectangle and the rate-pair taxonomy already operate). Belonging to the spine itself: the law/gauge split, the calibration of the two practices against each other on shared levels, and the vanishing hierarchy that falls out. Exact citations to be completed by the author.

10 · LEDGER

The spine's core identities, re-run in this page over randomized rigs at load time. The instruments and this ledger are evidence – randomized instances verified to floating-point noise; the proofs are the closed forms in their sources, and Theorem 0 is proved where it is stated.

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ok Reach gauge: equal steps [worst 4.3e-14]
ok Turn gauge: harmonic progression [worst 2nd-diff 1.1e-15]
ok calibration: shared level 0 and level 16 [worst 1.4e-14]
ok re-gauged marks collinear with the armature [worst 4.5e-13]
ok level  $\infty$  lands on  $h2 \cdot A \cdot h1$  [worst 3.5e-7]
ok v pair unreachable at any depth
ok geometric:  $Xg \cdot Xcs = x^2$  (the exchanger) [worst 1.8e-12]
ok arithmetic:  $x = AM(Xg, x^2/Xcs)$  [worst 2.8e-14]
ok harmonic:  $x = HM(Xcs, x^2/Xg)$  [worst 4.3e-14]
ok the anharmonic interface: cross-ratio invariant under Möbius maps
[worst 3.1e-15]
  ok sheet-39 law: straight lines on a Möbius gauge (gauge freedom) [worst
7.3e-12]

11/11 identities hold
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Spine v2 · supersedes v1 after audit: Theorem 0 stated; "graduation" split into law and gauge; §6 duality replaced by the rate-pair classification (the surviving fragment demoted to a flagged conjecture); the means stated as a conjugate pair; staging credited as calibration's precedent; ledger identity for cross-ratio invariance upgraded from affine to Möbius maps. Companion to the glide/cross-slit measuring board; relative links assume deployment beside [35g-tetra-drafting-board-glide.html](#) .

